

CELIA RUBIO-MADRIGAL

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 LinkedIn

 Google Scholar

I work on machine learning for graph and structured data; particularly, on how input structures shape data representation and computation, with broader interests in generalization trade-offs and architectural efficiency.

EXPERIENCE



- **Doctoral researcher** – CISPA Helmholtz Center for Information Security 2023-now
– Graph ML, efficiency, benchmarking, rewiring. – Advisor: Dr. Rebekka Burkholz  
- **RA: Departmental collaboration** – Ministerio de Educación & Universidad Complutense de Madrid 2021-22
– Identifying complex Boolean functions with ML  – Advisor: Dr. Ismael Rodríguez Laguna 
- **REU: Introduction to research ICMAT-Severo Ochoa** – Instituto de Ciencias Matemáticas (ICMAT) 2021
– “Differential geometry methods in ML applications”  – Advisor: Dr. David Martín de Diego 
- **Intern: Consultant Developer** – Guidewire Software – NLP search engine to find internal raised issues. 2021

EDUCATION



- **PhD in Computer Science** – CISPA & Universität des Saarlandes (Saarbrücken, Germany) 2023-now
- **MSc in Advanced CS with AI** – University of Strathclyde (Glasgow, UK) – 8.57/10 (Distinction) 2022-23
– Recipient of the “la Caixa” Fellowship  – Department’s Best Overall Performance Prize (Top 1)
- **BSc in Mathematics** – Universidad Complutense de Madrid (Spain) – 9.08/10 2017-22
- **BSc in Computer Science** – Universidad Complutense de Madrid (Spain) – 9.16/10 2017-22

PUBLICATIONS



Legend: **bold**: First-author A* conference. * Equal contribution. † Co-supervision.

-  R-M., Burkholz. *Linear GCNs Need Better Bias, Not More Expressive Power*. GFM Workshop @ ICML’26.
-  Joshi, Gadhikar†, R-M.†, Dutta, Singh, Burkholz. *Prune to Protect: Faster Training and Enhanced Privacy by Dynamic Data Pruning*. MemFM Workshop @ ICML’26.
-  R-M., Burkholz. *Fixed Aggregation Features Can Rival GNNs*. **ICML’26**.
-  Reddy, Verma, R-M., Muandet, Burkholz. *Boosting for Predictive Sufficiency*. ICLR’26.
-  Reddy, R-M., Burkholz, Muandet. *When Shift Happens - Confounding Is to Blame*. ICLR’26.
-  Jacobs, Gadhikar, R-M., Burkholz. *Hyperbolic Aware Minimization: Implicit Bias for Sparsity*. ICLR’26.
-  R-M.*, Jamadandi*, Burkholz. *GNNs Getting Comfy: Community and Feature Similarity Guided Rewiring*. **ICLR’25**.
-  Jamadandi*, R-M.*, Burkholz. *Spectral Graph Pruning against Over-squashing and Over-smoothing*. **NeurIPS’24**.

AWARDS



- 13th Heidelberg Laureate Forum: 200 international young researchers network with math and CS laureates 2026
- Dept. Best Overall Performance Prize: University of Strathclyde’s top student in MSc Advanced CS 2023
-  Nova III Student List: Spain’s top 10 talents under 25 in Computer Science 2023

- 9th Heidelberg Laureate Forum. Featured in: HLFF Spotlight, Tribuna Complutense 2022
- "la Caixa" Fellowship: Full funding for MSc studies. Featured in: El País, El Plural, TeleMadrid 2022
- **Undergraduate**
 - Aula de Verano "Blas Cabrera" for Spain's top undergrad records: Intl. Univ. Menéndez Pelayo (UIMP) 2022
 - SWERC contestant (competitive programming): ICPC Southwestern Europe Regional Contest in Italy 2022
 - Becas de Excelencia: Excellence Grants for undergrads with >9/10 in Madrid (2100€) 2018-22
- **Pre-undergraduate**
 - Premio Extraordinario de Bachillerato: Madrid's top 25 in general-content exam (1000€) 2017
 - EGMO contestant representing Spain: European Girls' Mathematical Olympiad in Switzerland 2017

TALKS



- **Invited talks** – Seminars and reading groups 2025-26
 - GLOW , TUWien ML RU seminar , Cohere For AI , CISPA ML seminar, AI Security Afternoon .
- **Contributed talks** – Conferences and workshops 2024-26
 - GSP 2026, HAICON 2024. Lightning talks: GTML 2025, MLG 2025 (2), NetSci 2025.
- **Non-archival posters** – Non-archival conferences, workshops, and sessions 2024-26
 - GSP 2026, CPAL 2026 (3), WiML 2024 2025, ELLIS Pre-NeurIPS/ICML 2024 2025 2026.

TEACHING



- **Math Olympiad Trainer** – International EGMO Preparatory Seminar in Bonn, Germany 2026
- **Master's and Bachelor's thesis advising** – CISPA & Universität des Saarlandes (3) 2024-now
- **Teaching Assistant: Laboratory demonstrator (7h/week)** – University of Strathclyde 2023
 - "CS310 Foundations of AI" for Year 3 BSc Computer Science – Lecturer: Dr. Andrew Abel

SERVICE



- **Academic service** 2024-now
 - Reviewer: MLG 2025-26, ICLR 2026 – **Top 200 (1%)**, CPAL 2026, ICML 2026 – Gold (25%), NeurIPS 2026.
 - Volunteer at 19th and 20th Women in Machine Learning Workshop (WiML) at NeurIPS 2024 and 2025.
- **Math Olympiad service**
 - IT at European Girls' Mathematical Olympiad (EGMO) with Joseph Samuel Myers 2023-now
 - Deputy Leader of Spain at EGMO 2022 and PAGMO 2021. 2021-22
 - Jury at the local phases of 57th to 61st Olimpiada Matemática Española (OME) Madrid. 2020-24
 - Guide at the Iberoamerican Mathematical Olympiad 2018.
- **Student representation** – Facultad de Informática (Fdi), Universidad Complutense de Madrid 2020-22
 - Secretary of the Fdi Student Council 2021-22. Member of the Fdi Faculty Board 2020-22.
- **L^AT_EX package development: ddphonism** 2019
 - Twelve-tone music notation diagrams with TikZ. Included in MiKTeX and TeXLive (try it in Overleaf).
- **Mentoring and outreach**
 - Mentor for new "la Caixa" Fellows 2023-25
 - Outreach articles for DivulgaMAT (RSME journal): Serialismo y matemáticas 2019
 - Outreach videos for the Spanish math education community: 2018-21